Multiple Liner regression

Multiple Linear Regression is a supervised machine learning algorithm used to predict a continuous numeric value when there are two or more independent variables (inputs) and one dependent variable (output).

It finds the relationship between multiple input variables and one output variable.

Jab ek output ko predict karne ke liye ek se zyada input variables use hote hain, tab Multiple Linear Regression use hota hai.

House Price Prediction

Hum predict karenge:

Inputs (X):

House_Size (sq ft)

Bedrooms

Age_of_House (years)

Output (y):

House_Price

Real life me house price sirf size se nahi decide hota.

Bedrooms aur house age bhi important hote hain.

Isliye Simple Linear Regression enough nahi hai

Yaha Multiple Linear Regression use karenge.

SIMPLE vs MULTIPLE LINEAR REGRESSION

Feature	Simple Linear Regression	Multiple Linear Regression
Input Variables	1	2 or more
Output Variable	1 numeric	1 numeric
Example	Study Hours → Marks	Size + Bedrooms + Age → Price

PART 5 – EQUATION

Simple Linear Regression Equation

$$y = mx + c$$

Multiple Linear Regression Equation

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3$$

Where:

- **y** = predicted output
- **b_0** = intercept
- **b_1, b_2, b_3** = coefficients
- **x_1, x_2, x_3** = input variables

Easy Meaning

- Har input variable ka output par **alag contribution** hota hai.

Example:

- Size badhe → price badhe
- Bedrooms badhe → price badhe
- House age badhe → price kam ho sakta hai

PART 6 – WHY MULTIPLE LINEAR REGRESSION?

Reason

Real world problems me usually:

❌ Sirf 1 factor ka effect nahi hota
✅ Multiple factors milkar output decide karte hain

Example:

House price depends on:

- size
- location
- bedrooms
- age
- floor
- parking

👉 Isliye Multiple Linear Regression practical world me bahot useful hai